RR-L-1829
August 17, 1973
SUPERSEDING
MIL-L-40610B
April 24, 1969

FEDERAL SPECIFICATION

LINER, BAKING CUP, PAPER

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE

1.1 This specification covers the requirements for treated paper liners of bleached chemical wood pulp for use as cup liners in bakeries (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

PPP-B-566	-	Boxes, Folding, Paperboard.
PPP-B-636	-	Boxes, Shipping, Fiberboard.
PPP-B-676	-	Boxes, Setup.
PPP-T-45	-	Tape, Gummed, Paper, Reinforced and Plain, for Sealing and Securing.

Federal Standard:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Speicifcations and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

FSC 7330

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(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-147 - Palletized Unit Loads for 40" x 48" Pallets.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Laws and Regulations:

21 CFR 121.2546 - Pulp from Reclaimed Fiber.

(The Code of Federal Regulations (CFR) and Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, US Government Printing Office, Washington, DC 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

Technical Association of the Pulp and Paper Industry Publications (TAPPI)

T 401 - Fiber Analysis of Paper and Paperboard.
T 402 - Standard Conditioning and Testing Atmospheres for Paper, Board, Pulp Handsheets and Related Products.
T 403 - Bursting Strength of Paper.
T 410 - Weight per Unit Area (Basis Weight or Substance) of Paper and Paperboard.
T 414 - Internal Tearing Resistance of Paper.

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(Application for copies should be addressed to the Technical Association of the Pulp and Paper Industry, 1 Dunwoody Park, Atlanta, GA 30341.)

National Motor Freight Traffic Association, Inc., Agent

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Material. The liner paper shall consist of 100 percent bleached chemical wood fibers and additives necessary to meet the requirements of this specification with the following exceptions: if reclaimed fibers are used, the liner paper may contain traces of fibers other than bleached chemical wood (not exceeding 5 percent.) Reclaimed fibers in any percentage may be used provided that materials, processes and finished liners conform to the specification and 21 CFR 121.2546. Reclaimed fibers, additives or after treatments shall not impart any odor or flavor to products baked in the liners.

3.2 Physical properties. The paper used in fabricating the liner shall conform to the requirements of table I when tested as specified in 4.2.1.1.

TABLE I. Physical requirements (minimum)

Basis weight, (24 x 36 - 500), pounds	28
Bursting strength, points	14
Tearing resistance, grams	
Machine direction	17
Cross direction	17

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3.3 Construction and dimensions.

3.3.1 Unless otherwise specified (see 6.2), liners shall be furnished in the form of circles having a diameter of 4-1/2 inches ± 1/8 inch and uniformly fluted to form a cup having a 2 ± 1/16 inch, circular bottom with fluted slanting sidewalls having a slant height of 1-1/4 ± 1/16 inches.

3.4 Performance. The liners shall release easily from the pan without the necessity for pan greasing. The liner shall not adhere to the baked item to such extent that significant portions are pulled away when the liner is removed. This is not intended to include the occurrence of thin layers of crumbs which usually adhere to the liner.

3.5 Workmanship. The liners shall be free from any dirt, holes, tears, cuts, foreign matter, or undefibered stock and shall have clean-cut edges. The liners shall conform to the levels of quality established herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification, where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification. The supplier shall furnish to the contracting officer a certificate of compliance to the effect that all additives or after treatments used in the liners conform in all respects to 21 CFR 121.2546 (see 3.1), that the liners impart no odor or flavor to products baked in them, and that the liners conform to the release property (performance) requirements (see 3.4).

4.2 Quality conformance inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.2.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be tested in accordance with all the requirements of referenced specifications, and standards unless otherwise excluded, amended, or qualified in this specification or applicable purchase documents.

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4.2.1.1 Testing of the component material. The paper used in the fabrication of the liners shall be conditioned in accordance with TAPPI method T 402 and tested in accordance with the applicable TAPPI test method specified in table II. The lot size for purposes of determining the sample size shall be expressed in units of pounds of paper to be used for the fabrication of the liner as received at one time from one source of supply. The sample unit shall be 1,000 square inches of paper. The sample size shall be drawn at inspection level S-1 of MIL-STD-105. There shall be no failure to meet the composite or sample unit requirements. Test reports shall include all values upon which the results are based. A composite shall consist of small portions taken from all sample units and combined in a single sample.

TABLE II. Instructions for testing of components

Characteristic	Specification reference		Requirements applicable to		Number determinations per sample unit	Results reported as	
	Requirement	Test method	Individual unit	Lot average		Pass or fail	Numerically to nearest
Stock	3.1	T 401		X	Average of 2 on composite		5 percent
Basis weight	3.2	T 410	X		1		0.1 pound
Bursting strength	3.2	T 403	X		Average of 10 (5 on each side)		point
Tearing resistance							
Machine direction	3.2	T 414	X		Average of 10		gram
Cross direction	3.2	T 414	X		Average of 10		gram

4.3 Inspection of the end item.

4.3.1 Examination of the end item. The end item shall be examined for the defects set forth in the applicable subparagraphs at the inspection levels and acceptable quality levels (AQLs) set forth in 4.3.1.6. Random samples shall be drawn from each lot of the end item for examination of visual, dimensional, count, preparation for delivery, and palletization defects. The

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lot size for purposes of determining the sample size in accordance with MIL-STD-105 shall be expressed in units of unit packages for examination in 4.3.1.1, 4.3.1.2, and 4.3.1.3; in units of shipping containers for examination in 4.3.1.4; and in units of palletized unit loads for examination in 4.3.1.5.

4.3.1.1 Examination for visual defects. The sample unit for this examination shall be one liner. Both sides of the liner shall be examined.

Examine	Defects
Appearance	Not clean Any foreign matter or undefibered stock Any tear, hole, or cut Any wrinkle, crease, or fold not consistent with construction requirements
Workmanship	Baking cup liner not uniformly fluted Misshapen Ragged or uneven edges

4.3.1.2 Examination for defects in dimension. The sample unit for this examination shall be one liner.

Examine	Defects
Baking cup liner	
Overall diameter	Less than 4 inches or more than 5 inches
Diameter of base	Less than 1-15/16 inches or more than 2-1/16 inches
Sidewall height (slant)	Less than 1-3/16 inches or more than 1-5/16 inches

4.3.1.3 Examination for count. The sample unit for this examination shall be one unit package. The lot shall be unacceptable if the average count per package for all packages examined is less than specified or indicated quantity.

4.3.1.4 Examination of preparation for delivery requirements. An examination shall be made to determine that the packaging, packing, and marking complies with the section 5 requirements. Defects shall be scored in accordance with the list below. The sample unit for this examination shall be one shipping container fully prepared for delivery except that it shall not be closed. Defects of closure listed below shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot.

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Examine	Defects
Markings (exterior and interior)	Omitted; incorrect; incomplete; illegible; of improper size, location, sequence, or method of application.
Materials	Any component missing, damaged, or otherwise defective. Any component not as specified.
Workmanship	Inadequate application of components such as incomplete closure of container flaps; loose strapping, improper taping, or inadequate stapling. Bulged or distorted container.
Contents	Number of liners per shipping container more or less than specified. Gross weight exceeds requirements of box specification.

4.3.1.5 Examination for palletization. An examination shall be made to determine that the palletization complies with the section 5 requirements. Defects shall be scored in accordance with the list below. The sample unit shall be one palletized unit load fully prepared for delivery. The lot size shall be the number of palletized unit loads in the end item inspection lot.

Examine	Defect
Finished dimension	Length, width, or height exceeds specified maximum requirement.
Palletization	Not as specified. Pallet pattern not as specified. Interlocking of loads not as specified. Load not bonded with required straps as specified.
Weight	Exceeds maximum load limits.
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.

4.3.1.6 Inspection levels and AQLs for examinations. The inspection levels, for purposes of determining the sample size, and the AQLs expressed in defects per one hundred units shall be as follows:

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Examination paragraph	Inspection level	AQL
4.3.1.1	II	2.5
4.3.1.2	I	4.0
4.3.1.3	S-1	N.A.
4.3.1.4	S-1	2.5
4.3.1.5	S-1	6.5

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B, or C as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Unit packaging. Five hundred liners of one description only shall be unit packaged in a box conforming to optional classification of PPP-B-566 or PPP-B-676. Each paperboard box shall be securely closed in accordance with the appendix of the applicable box specification.

5.1.1.2 Intermediate packaging. Ten thousand liners of one description only, unit packaged as specified in 5.1.1.1, shall be packaged in a fiberboard box conforming to type CF (variety SW), class domestic, grade 200 of PPP-B-636. Each box shall be securely closed with 2-inch minimum width gummed paper tape conforming to type III, grade B of PPP-T-45.

5.1.2 Level B. Five hundred liners of one description only shall be unit packaged in a commercial type paperboard tube. The tube shall be securely closed in a manner to prevent accidental opening and to protect the contents from contamination. The liners shall then be further packaged as specified in 5.1.1.2.

5.1.3 Level C. Liners shall be packaged to afford adequate protection against physical damage during shipment from the supply source to the first receiving activity. The supplier may use his standard practice when it meets this requirement.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. Sixty thousand liners of one description only, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, grade V2s of PPP-B-636. Each shipping container shall be closed in accordance with method III, waterproofed in accordance with method V, and reinforced as specified in the appendix of the container specification.

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5.2.2 Level B. Sixty thousand liners of one description only, packaged as specified in 5.1, shall be packed in a fiberboard shipping container conforming to type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636. Each shipping container shall be closed in accordance with method II as specified in the appendix of the container specification.

5.2.2.1 When specified (see 6.2), the fiberboard shipping container shall be a grade V3c, V3s, or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with method III as specified in the appendix of the container specification.

5.2.3 Level C. Liners, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable.

5.3 Palletization. When specified (see 6.2), liners of one description only, packed as specified in 5.2, shall be palletized in accordance with load type I of MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means K and L. Pallet patterns shall be in accordance with the appendix of MIL-STD-147. Interlocking of loads shall be effected by reversing the pattern of each course. If the container is of a size which does not conform to any of the pallet patterns specified in MIL-STD-147, the pallet pattern used shall first be approved by the contracting officer.

5.4 Marking. In addition to any special marking required by the contract or order, all markings shall be in accordance with 5.4.1 or 5.4.2, as specified (see 6.2).

5.4.1 Civil agencies. Marking shall be in accordance with Fed. Std. No. 123.

5.4.2 Military requirements. Marking shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Cup liners covered by this specification are used with pans for baking cup cakes to eliminate the need for greasing and excessive cleaning of pans.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Dimensions of liners when other than specified (3.3.1).

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(c) Selection of the applicable levels of packaging and packing (see 5.1 and 5.2).
(d) Whether marking shall be in accordance with Civil agencies or Military requirements (see 5.4).
(e) When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).
(f) When palletization is required (see 5.3).

MILITARY INTERESTS:

Custodians:

Army - GL
Navy - SA
Air Force - 84

Review activities:

Army - MD
Navy - MS

User activities:

Navy - MC, CG

Preparing activity:

Army - GL

Civil Agency Coordinating Activities:

DC Govt - DCG
GSA - FSS
HEW - HSM
VA - DMS

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

RR-L-1829 - Liner, Baking Cup, Paper

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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